Praneeth Mangala

Praneethmangala339@gmail.com | +91 8328575111 | Portfolio | GitHub | LinkedIn

Research Experience

Research Intern (Remote) — NIT Raipur (Prof. Dr. V. K. Vidyarthi) | Aceptance Mail Jan '26 – Present

- **Scientific ML:**Developing PINNs in PyTorch to solve nonlinear hydrological **PDEs**& subsurface flow via **Richards equation**
- **Optimization:** Embedding physical laws into custom loss functions, enhancing stability & convergence over data-only models
- **Scalability:** Bridging numerical methods with deep learning to build scalable, physics-guided models for fluid dynamics

Final Year Research Thesis — VNIT Nagpur (Prof. Dr. Anamika Singh) Sep '24 – Apr '25

- Developed a guided-medium communication testbed achieving N-bit transmission with a Bit Error Rate (**BER**) of **<5%**
- Engineered an N-bit transmission algorithm using solenoid valves and **FOOK** modulation to optimize fluidic signaling
- **Presented** research at **ICMOCE'25 (IIT BBS)** and **NCC'25 (IIT Delhi)**; filed a patent for the novel hardware architecture

Work Experience

Agentflow.ai Systems — Founder | Bengaluru, India Jul 2026 – Present

- Building **Memory OS**, a governance layer for agentic AI, controlling agent memory, permissions, access, and auditability
- Designing **Agentflow.ai Systems**, a runtime for orchestrating AI agents with memory, tools, workflows, and governance controls

Orbis AI Technologies — AI Lead Engineer | Bengaluru, Karnataka Aug'26 – Present

- Leading development of **FactoryOS V1**, an AI-native manufacturing operations platform
- Building scalable frontend and backend using **React, FastAPI, PostgreSQL, and AWS**
- Designing factory knowledge and context layers across machines, production, workforce, quality, and maintenance
- Developing **RAG and memory systems** for SOPs, manuals, and operational knowledge
- Building AI copilots with **human approval, explainability, and audit logging**

Eka Mobility (R&D Software Engineer) Pune,Maharashtra July '25 – Jan'26

- Analyzed datasets to characterize demand via time series analysis, identifying hourly, weekly, and monthly patterns
- Performed time series analysis on 8-month telemetry to estimate SoC and SoH, optimizing life cycle accuracy by **15%**
- **Predictive Modeling:** Developed Scikit-learn models to identify operational bottlenecks, reducing system-wide errors by **20%**
- Built SQL pipelines for **100k+ daily records**, optimizing indexing to improve query response times by **30%**
- **Frontend Dev:** Architected the **EKA-Connect** React interface to deliver interactive dashboards for real-time business insights

Education

Visvesvaraya National Institute of Technology (VNIT), Nagpur Oct '21 – May '25
Bachelor of Technology in Electronics and Communication Engineering | **CGPA: 3.09/4**

Shastra Junior College June '19 – May '21
Higher Secondary Certificate (HSC) (Mathematics , Physics , Chemistry | *Class XII* | **Percentage: 98.6%**

Relevant Coursework

- **Mathematics:** Linear Algebra, Numerical Linear Algebra, Numerical Methods and Probability Theory & Statistics, Calculus
- **Computer Science:** Data Structures, Algorithms, Operating Systems, Object Oriented Programming (OOP), C Language
- **Applied ML:** Machine Learning in Python, Physics-Informed ML, Deep Learning, Digital Signal Processing, Embedded Systems

Projects

1D Simple Harmonic Oscillator | *PyTorch, Python, NumPy, Matplotlib* | GitHub Link

- Engineered a **PyTorch-based PINN** to simulate 1D underdamped oscillators by integrating ODE constraints ($d < \omega_0$) into a custom loss function allowing model to learn complex dynamics by embedding physical laws directly into neural architecture
- Optimized a multi-layer FCN with Tanh activations and dual-objective strategy to minimize data and physics residuals ensuring predictions remained physically consistent while maintaining high-fidelity accuracy throughout the training process

- Built an automated pipeline to track convergence against analytical solutions and generated training GIFs to verify adherence to physics providing a robust framework for real-time monitoring of model integrity and predictive performance

App Graph Builder | *React, TypeScript, Vite, ReactFlow, Zustand, TanStack Query* | GitHub Link

- Engineered ReactFlow-based visual node system for building modular application workflows with drag&drop execution flow
- Designed a Zustand-powered state graph engine managing node lifecycle,inter-node communication,runtime data propagation
- Built a scalable architecture with pluggable node types (Input, LLM, API) enabling future agentic workflow orchestration systems

AI Co-Founder System | *Agentic AI, RAG, LLM Orchestration, Python* | GitHub Link

- Built agentic AI co-founder system converting startup ideas into execution plans using multi-agent reasoning
- Designed Planner, Validator, Executor agents for feasibility, decomposition, and strategy generation
- Implemented Cognee AI + RAG memory enabling contextual reasoning and iterative decision refinement

StayEase (PG Management System) | *Flutter, Firebase, Firestore, Node.js*

- Built a full-stack PG management system enabling real-time tenant tracking,rent collection, multi-floor operational monitoring
- Designed role-based admin system supporting owner/tenant views with access control, notifications, and complaint management
- Implemented real-time Firebase Firestore sync for tenants, payments, and occupancy enabling live cross-device updates
- Developed modular UI for onboarding, room allocation, and payment tracking improving scalability across properties
- Architected system with future hooks for AI-driven rent prediction, complaint classification, and occupancy optimization

Analog to Digital Converter | *8086 Assembly, Hardware Interfacing, Embedded C* | GitHub Link

- Engineered an 8086 system to interface with ADC/DAC components using the **8255 PPI** and **8253 Timer** to enable synchronized acquisition allowing for precise hardware-level control and real-time data synchronization across external peripherals
- Developed a sampling routine for 150 points and implemented a 32-point moving average filter using SAR (Arithmetic Shift Right) instructions optimizing computational efficiency by replacing expensive division with bit-shifting for real-time noise reduction.
- Designed a continuous processing loop to reproduce stabilized signals through a DAC, verifying low-level hardware-software synchronization demonstrating high-fidelity signal reconstruction and robust embedded system performance

Achievements & Honors

- Among **Top 1%** in JEE Mains 2021 (1.2M candidates)
- Selected for research demonstration at **NCC'25 (IIT Delhi)** and presented thesis at **ICMOCE'25 (IIT BBS)** | Certificate
- Awarded **$6,000** (CBI Cybersecurity Hackathon) and **2nd Place** at SEED (**$420** prize + 50% waiver) | Award Proof
- Global Finalist at **GFF'25** for FraudLens and **Runner-up** at BITS Pilani Innovators Conclave'25 | Certificate
- Achieved a perfect **GPA of 10/10** in 10th Grade for exceptional overall academic performance | Score Card
- Runner-up,RBI Global Hackathon–HaRBInger'25 for AI-driven digital payments trust platform, securing **$30K** in funding/support
- Awarded **$600** and Jury Award at HackIITK Cybersecurity Hackathon for an AI-driven fraud detection platformCertificate

Technical Skills

- **Languages:** Python, C/C++, SQL , 8086 Assembly, MATLAB, JavaScript (ES6+), HTML5/CSS3.
- **Libraries:** PyTorch, NumPy, SciPy, Scikit-learn, Matplotlib, Pandas, Statsmodels (Time Series Analysis)
- **Frameworks:** React.js , Physics-Informed Neural Networks (PINNs), Express.js, Node.js, REST APIs.
- **Developer Tools:** VS Code,Jupyter Notebook, Git/GitHub, LaTeX

Leadership & Activities

- **Technical Upskilling:** Participated in advanced workshops including **AI for Wireless Communications**, focusing on signal processing and ML by *Prof. Arumugam Nallanathan (Queensmary University , London)* | Certificate.
- **Event Manager (AXIS'24):** Led logistics for VNIT's national techfest; managed 20+ volunteers to execute high-stakes competitive events | LoR.
- **Creative Secretary (ECE Dept):** Spearheaded branding & design strategy, leading department to **2nd Place** finish in IG'24